MSG No. 227.129 $Fd\bar{3}m1'$ [Type II, cubic]

Table 1: Wyckoff site: 8a, site symmetry: -43m1'

No.	position	mapping
1	$\left[\frac{1}{8}, \frac{1}{8}, \frac{1}{8}\right]$	[1,8,9,10,17,18,19,20,21,22,23,24,26,27,28,29,30,31,35,36,37,38,39,40,
		193,200,201,202,209,210,211,212,213,214,215,216,218,219,220,221,222,223,227,228,229,230,231,232]
2	$\left[\frac{3}{8}, \frac{7}{8}, \frac{3}{8}\right]$	[2,7,15,33,43,48,51,52,59,82,92,93,108,110,112,121,137,138,149,150,157,176,190,191,
		194,199,207,225,235,240,243,244,251,274,284,285,300,302,304,313,329,330,341,342,349,368,382,383]
3	$\left[\frac{3}{8}, \frac{3}{8}, \frac{7}{8}\right]$	[3,4,11,34,44,45,50,55,63,81,91,96,101,102,109,128,142,143,156,158,160,169,185,186,
		195,196,203,226,236,237,242,247,255,273,283,288,293,294,301,320,334,335,348,350,352,361,377,378]
4	$\left[\frac{7}{8}, \frac{3}{8}, \frac{3}{8}\right]$	[5,6,13,32,46,47,60,62,64,73,89,90,99,100,107,130,140,141,146,151,159,177,187,192,
		197,198,205,224,238,239,252,254,256,265,281,282,291,292,299,322,332,333,338,343,351,369,379,384]
5	$\left[\frac{7}{8}, \frac{7}{8}, \frac{7}{8}\right]$	[12,14,16,25,41,42,53,54,61,80,94,95,98,103,111,129,139,144,147,148,155,178,188,189,
		204,206,208,217,233,234,245,246,253,272,286,287,290,295,303,321,331,336,339,340,347,370,380,381]
6	$\left[\frac{1}{8}, \frac{5}{8}, \frac{5}{8}\right]$	[49,56,57,58,65,66,67,68,69,70,71,72,74,75,76,77,78,79,83,84,85,86,87,88,
		241,248,249,250,257,258,259,260,261,262,263,264,266,267,268,269,270,271,275,276,277,278,279,280]
7	$\left[\frac{5}{8}, \frac{1}{8}, \frac{5}{8}\right]$	[97,104,105,106,113,114,115,116,117,118,119,120,122,123,124,125,126,127,131,132,133,134,135,136,
		289,296,297,298,305,306,307,308,309,310,311,312,314,315,316,317,318,319,323,324,325,326,327,328]
8	$\left[\frac{5}{8}, \frac{5}{8}, \frac{1}{8}\right]$	[145,152,153,154,161,162,163,164,165,166,167,168,170,171,172,173,174,175,179,180,181,182,183,184,
		337,344,345,346,353,354,355,356,357,358,359,360,362,363,364,365,366,367,371,372,373,374,375,376]

Table 2: Wyckoff site: 8b, site symmetry: -43m1'

No.	position	mapping
1	$\left[\frac{3}{8}, \frac{3}{8}, \frac{3}{8}\right]$	[1,17,18,36,38,40,56,70,71,77,78,85,105,115,120,122,127,135,154,164,165,171,172,179,
		193,209,210,228,230,232,248,262,263,269,270,277,297,307,312,314,319,327,346,356,357,363,364,371]
2	$\left[\frac{5}{8}, \frac{5}{8}, \frac{5}{8}\right]$	[2,3,4,5,6,7,11,12,13,14,15,16,25,32,33,34,41,42,43,44,45,46,47,48,
		194,195,196,197,198,199,203,204,205,206,207,208,217,224,225,226,233,234,235,236,237,238,239,240]
3	$\left[\frac{3}{8}, \frac{7}{8}, \frac{7}{8}\right]$	[8,22,23,29,30,37,49,65,66,84,86,88,106,116,117,123,124,131,153,163,168,170,175,183,
		200,214,215,221,222,229,241,257,258,276,278,280,298,308,309,315,316,323,345,355,360,362,367,375]
4	$\left[\frac{7}{8}, \frac{3}{8}, \frac{7}{8}\right]$	[9,19,24,26,31,39,58,68,69,75,76,83,97,113,114,132,134,136,152,166,167,173,174,181,
		201,211,216,218,223,231,250,260,261,267,268,275,289,305,306,324,326,328,344,358,359,365,366,373]
5	$\left[\frac{7}{8}, \frac{7}{8}, \frac{3}{8}\right]$	[10,20,21,27,28,35,57,67,72,74,79,87,104,118,119,125,126,133,145,161,162,180,182,184,
		202,212,213,219,220,227,249,259,264,266,271,279,296,310,311,317,318,325,337,353,354,372,374,376]
6	$\left[\frac{5}{8}, \frac{1}{8}, \frac{1}{8}\right]$	[50,51,52,53,54,55,59,60,61,62,63,64,73,80,81,82,89,90,91,92,93,94,95,96,
		242,243,244,245,246,247,251,252,253,254,255,256,265,272,273,274,281,282,283,284,285,286,287,288]
7	$\left[\frac{1}{8}, \frac{5}{8}, \frac{1}{8}\right]$	[98,99,100,101,102,103,107,108,109,110,111,112,121,128,129,130,137,138,139,140,141,142,143,144,
		290,291,292,293,294,295,299,300,301,302,303,304,313,320,321,322,329,330,331,332,333,334,335,336]
8	$\left[\frac{1}{8}, \frac{1}{8}, \frac{5}{8}\right]$	[146,147,148,149,150,151,155,156,157,158,159,160,169,176,177,178,185,186,187,188,189,190,191,192,
		338,339,340,341,342,343,347,348,349,350,351,352,361,368,369,370,377,378,379,380,381,382,383,384]

Table 3: Wyckoff site: 16c, site symmetry: .-3m1'

No.	position	mapping
1	[0, 0, 0]	[1,12,14,16,17,18,25,36,38,40,41,42,193,204,206,208,209,210,217,228,230,232,233,234]
2	$\left[\frac{1}{4}, 0, \frac{1}{4}\right]$	[2,7,9,15,19,24,26,31,33,39,43,48,194,199,201,207,211,216,218,223,225,231,235,240]

continued ...

Table 3

No.	position	mapping
3	$\left[\frac{1}{4}, \frac{1}{4}, 0\right]$	[3,4,10,11,20,21,27,28,34,35,44,45,195,196,202,203,212,213,219,220,226,227,236,237]
4	$\left[0, \frac{1}{4}, \frac{1}{4}\right]$	[5,6,8,13,22,23,29,30,32,37,46,47,197,198,200,205,214,215,221,222,224,229,238,239]
5	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	[49,60,62,64,65,66,73,84,86,88,89,90,241,252,254,256,257,258,265,276,278,280,281,282]
6	$\left[\frac{1}{4}, \frac{1}{2}, \frac{3}{4}\right]$	[50,55,57,63,67,72,74,79,81,87,91,96,242,247,249,255,259,264,266,271,273,279,283,288]
7	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2}\right]$	[51,52,58,59,68,69,75,76,82,83,92,93,243,244,250,251,260,261,267,268,274,275,284,285]
8	$\left[0, \frac{3}{4}, \frac{3}{4}\right]$	[53,54,56,61,70,71,77,78,80,85,94,95,245,246,248,253,262,263,269,270,272,277,286,287]
9	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	[97,108,110,112,113,114,121,132,134,136,137,138,289,300,302,304,305,306,313,324,326,328,329,330]
10	$\left[\frac{3}{4}, 0, \frac{3}{4}\right]$	[98,103,105,111,115,120,122,127,129,135,139,144,290,295,297,303,307,312,314,319,321,327,331,336]
11	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}\right]$	[99,100,106,107,116,117,123,124,130,131,140,141,291,292,298,299,308,309,315,316,322,323,332,333]
12	$\left[\frac{1}{2}, \frac{1}{4}, \frac{3}{4}\right]$	[101,102,104,109,118,119,125,126,128,133,142,143,293,294,296,301,310,311,317,318,320,325,334,335]
13	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	[145,156,158,160,161,162,169,180,182,184,185,186,337,348,350,352,353,354,361,372,374,376,377,378]
14	$\left[\frac{3}{4}, \frac{1}{2}, \frac{1}{4}\right]$	[146,151,153,159,163,168,170,175,177,183,187,192,338,343,345,351,355,360,362,367,369,375,379,384]
15	$\left[\frac{3}{4}, \frac{3}{4}, 0\right]$	[147,148,154,155,164,165,171,172,178,179,188,189,339,340,346,347,356,357,363,364,370,371,380,381]
16	$\left[\frac{1}{2}, \frac{3}{4}, \frac{1}{4}\right]$	[149,150,152,157,166,167,173,174,176,181,190,191,341,342,344,349,358,359,365,366,368,373,382,383]

Table 4: Wyckoff site: 16d, site symmetry: .-3m1'

No.	position	mapping
1	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	[1,12,14,16,17,18,25,36,38,40,41,42,193,204,206,208,209,210,217,228,230,232,233,234]
2	$\left[\frac{3}{4}, \frac{1}{2}, \frac{3}{4}\right]$	[2,7,9,15,19,24,26,31,33,39,43,48,194,199,201,207,211,216,218,223,225,231,235,240]
3	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{2}\right]$	[3,4,10,11,20,21,27,28,34,35,44,45,195,196,202,203,212,213,219,220,226,227,236,237]
4	$\left[\frac{1}{2}, \frac{3}{4}, \frac{3}{4}\right]$	[5,6,8,13,22,23,29,30,32,37,46,47,197,198,200,205,214,215,221,222,224,229,238,239]
5	$\left[\frac{1}{2}, 0, 0\right]$	[49,60,62,64,65,66,73,84,86,88,89,90,241,252,254,256,257,258,265,276,278,280,281,282]
6	$\left[\frac{3}{4}, 0, \frac{1}{4}\right]$	[50,55,57,63,67,72,74,79,81,87,91,96,242,247,249,255,259,264,266,271,273,279,283,288]
7	$\left[\frac{3}{4}, \frac{1}{4}, 0\right]$	[51,52,58,59,68,69,75,76,82,83,92,93,243,244,250,251,260,261,267,268,274,275,284,285]
8	$\left[\frac{1}{2}, \frac{1}{4}, \frac{1}{4}\right]$	[53,54,56,61,70,71,77,78,80,85,94,95,245,246,248,253,262,263,269,270,272,277,286,287]
9	$\left[0, \frac{1}{2}, 0\right]$	[97,108,110,112,113,114,121,132,134,136,137,138,289,300,302,304,305,306,313,324,326,328,329,330]
10	$\left[\frac{1}{4}, \frac{1}{2}, \frac{1}{4}\right]$	[98,103,105,111,115,120,122,127,129,135,139,144,290,295,297,303,307,312,314,319,321,327,331,336]
11	$\left[\frac{1}{4}, \frac{3}{4}, 0\right]$	[99,100,106,107,116,117,123,124,130,131,140,141,291,292,298,299,308,309,315,316,322,323,332,333]
12	$\left[0, \frac{3}{4}, \frac{1}{4}\right]$	[101,102,104,109,118,119,125,126,128,133,142,143,293,294,296,301,310,311,317,318,320,325,334,335]
13	$\left[0, 0, \frac{1}{2}\right]$	[145,156,158,160,161,162,169,180,182,184,185,186,337,348,350,352,353,354,361,372,374,376,377,378]
14	$\left[\frac{1}{4}, 0, \frac{3}{4}\right]$	[146,151,153,159,163,168,170,175,177,183,187,192,338,343,345,351,355,360,362,367,369,375,379,384]
15	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{2}\right]$	[147,148,154,155,164,165,171,172,178,179,188,189,339,340,346,347,356,357,363,364,370,371,380,381]
16	$\left[0, \frac{1}{4}, \frac{3}{4}\right]$	[149,150,152,157,166,167,173,174,176,181,190,191,341,342,344,349,358,359,365,366,368,373,382,383]

Table 5: Wyckoff site: 32e, site symmetry: .3m1'

No.	position	mapping
1	$[x, x, x]$	[1,17,18,36,38,40,193,209,210,228,230,232]
2	$\left[x+\frac{1}{4}, -x, x+\frac{1}{4}\right]$	[2,7,15,33,43,48,194,199,207,225,235,240]
3	$\left[x+\frac{1}{4}, x+\frac{1}{4}, -x\right]$	[3,4,11,34,44,45,195,196,203,226,236,237]
4	$\left[-x, x+\frac{1}{4}, x+\frac{1}{4}\right]$	[5,6,13,32,46,47,197,198,205,224,238,239]
5	$\left[x, \frac{1}{4}-x, \frac{1}{4}-x\right]$	[8,22,23,29,30,37,200,214,215,221,222,229]
6	$\left[\frac{1}{4}-x, x, \frac{1}{4}-x\right]$	[9,19,24,26,31,39,201,211,216,218,223,231]

continued ...

Table 5

No.	position	mapping
7	$\left[\frac{1}{4}-x,\ \frac{1}{4}-x,\ x\right]$	[10,20,21,27,28,35,202,212,213,219,220,227]
8	$[-x,\ -x,\ -x]$	[12,14,16,25,41,42,204,206,208,217,233,234]
9	$\left[x,\ x+\frac{1}{2},\ x+\frac{1}{2}\right]$	[49,65,66,84,86,88,241,257,258,276,278,280]
10	$\left[x+\frac{1}{4},\ \frac{1}{2}-x,\ x+\frac{3}{4}\right]$	[50,55,63,81,91,96,242,247,255,273,283,288]
11	$\left[x+\frac{1}{4},\ x+\frac{3}{4},\ \frac{1}{2}-x\right]$	[51,52,59,82,92,93,243,244,251,274,284,285]
12	$\left[-x,\ x+\frac{3}{4},\ x+\frac{3}{4}\right]$	[53,54,61,80,94,95,245,246,253,272,286,287]
13	$\left[x,\ \frac{3}{4}-x,\ \frac{3}{4}-x\right]$	[56,70,71,77,78,85,248,262,263,269,270,277]
14	$\left[\frac{1}{4}-x,\ x+\frac{1}{2},\ \frac{3}{4}-x\right]$	[57,67,72,74,79,87,249,259,264,266,271,279]
15	$\left[\frac{1}{4}-x,\ \frac{3}{4}-x,\ x+\frac{1}{2}\right]$	[58,68,69,75,76,83,250,260,261,267,268,275]
16	$\left[-x,\ \frac{1}{2}-x,\ \frac{1}{2}-x\right]$	[60,62,64,73,89,90,252,254,256,265,281,282]
17	$\left[x+\frac{1}{2},\ x,\ x+\frac{1}{2}\right]$	[97,113,114,132,134,136,289,305,306,324,326,328]
18	$\left[x+\frac{3}{4},\ -x,\ x+\frac{3}{4}\right]$	[98,103,111,129,139,144,290,295,303,321,331,336]
19	$\left[x+\frac{3}{4},\ x+\frac{1}{4},\ \frac{1}{2}-x\right]$	[99,100,107,130,140,141,291,292,299,322,332,333]
20	$\left[\frac{1}{2}-x,\ x+\frac{1}{4},\ x+\frac{3}{4}\right]$	[101,102,109,128,142,143,293,294,301,320,334,335]
21	$\left[x+\frac{1}{2},\ \frac{1}{4}-x,\ \frac{3}{4}-x\right]$	[104,118,119,125,126,133,296,310,311,317,318,325]
22	$\left[\frac{3}{4}-x,\ x,\ \frac{3}{4}-x\right]$	[105,115,120,122,127,135,297,307,312,314,319,327]
23	$\left[\frac{3}{4}-x,\ \frac{1}{4}-x,\ x+\frac{1}{2}\right]$	[106,116,117,123,124,131,298,308,309,315,316,323]
24	$\left[\frac{1}{2}-x,\ -x,\ \frac{1}{2}-x\right]$	[108,110,112,121,137,138,300,302,304,313,329,330]
25	$\left[x+\frac{1}{2},\ x+\frac{1}{2},\ x\right]$	[145,161,162,180,182,184,337,353,354,372,374,376]
26	$\left[x+\frac{3}{4},\ \frac{1}{2}-x,\ x+\frac{1}{4}\right]$	[146,151,159,177,187,192,338,343,351,369,379,384]
27	$\left[x+\frac{3}{4},\ x+\frac{3}{4},\ -x\right]$	[147,148,155,178,188,189,339,340,347,370,380,381]
28	$\left[\frac{1}{2}-x,\ x+\frac{3}{4},\ x+\frac{1}{4}\right]$	[149,150,157,176,190,191,341,342,349,368,382,383]
29	$\left[x+\frac{1}{2},\ \frac{3}{4}-x,\ \frac{1}{4}-x\right]$	[152,166,167,173,174,181,344,358,359,365,366,373]
30	$\left[\frac{3}{4}-x,\ x+\frac{1}{2},\ \frac{1}{4}-x\right]$	[153,163,168,170,175,183,345,355,360,362,367,375]
31	$\left[\frac{3}{4}-x,\ \frac{3}{4}-x,\ x\right]$	[154,164,165,171,172,179,346,356,357,363,364,371]
32	$\left[\frac{1}{2}-x,\ \frac{1}{2}-x,\ -x\right]$	[156,158,160,169,185,186,348,350,352,361,377,378]

Table 6: Wyckoff site: 48f, site symmetry: 2.mm1'

No.	position	mapping
1	$\left[x,\ \frac{1}{8},\ \frac{1}{8}\right]$	[1,8,37,38,193,200,229,230]
2	$\left[x+\frac{1}{4},\ \frac{7}{8},\ \frac{3}{8}\right]$	[2,33,51,82,194,225,243,274]
3	$\left[x+\frac{1}{4},\ \frac{3}{8},\ \frac{7}{8}\right]$	[3,34,50,81,195,226,242,273]
4	$\left[\frac{3}{8},\ \frac{3}{8},\ -x\right]$	[4,45,160,186,196,237,352,378]
5	$\left[\frac{7}{8},\ \frac{3}{8},\ x+\frac{1}{4}\right]$	[5,47,159,187,197,239,351,379]
6	$\left[\frac{7}{8},\ x+\frac{1}{4},\ \frac{3}{8}\right]$	[6,46,107,140,198,238,299,332]
7	$\left[\frac{3}{8},\ -x,\ \frac{3}{8}\right]$	[7,48,108,137,199,240,300,329]
8	$\left[\frac{1}{4}-x,\ \frac{1}{8},\ \frac{1}{8}\right]$	[9,10,26,27,201,202,218,219]
9	$\left[\frac{3}{8},\ x+\frac{1}{4},\ \frac{7}{8}\right]$	[11,44,102,142,203,236,294,334]
10	$\left[\frac{7}{8},\ -x,\ \frac{7}{8}\right]$	[12,41,103,144,204,233,295,336]
11	$\left[-x,\ \frac{3}{8},\ \frac{3}{8}\right]$	[13,32,62,73,205,224,254,265]
12	$\left[-x,\ \frac{7}{8},\ \frac{7}{8}\right]$	[14,25,61,80,206,217,253,272]
13	$\left[\frac{3}{8},\ \frac{7}{8},\ x+\frac{1}{4}\right]$	[15,43,149,191,207,235,341,383]
14	$\left[\frac{7}{8},\ \frac{7}{8},\ -x\right]$	[16,42,148,189,208,234,340,381]
15	$\left[\frac{1}{8},\ x,\ \frac{1}{8}\right]$	[17,24,31,36,209,216,223,228]

continued ...

Table 6

No.	position	mapping
16	$\left[\frac{1}{8}, \frac{1}{8}, x\right]$	[18,21,28,40,210,213,220,232]
17	$\left[\frac{1}{8}, \frac{1}{8}, \frac{1}{4}-x\right]$	[19,23,29,39,211,215,221,231]
18	$\left[\frac{1}{8}, \frac{1}{4}-x, \frac{1}{8}\right]$	[20,22,30,35,212,214,222,227]
19	$\left[x, \frac{5}{8}, \frac{5}{8}\right]$	[49,56,85,86,241,248,277,278]
20	$\left[\frac{3}{8}, \frac{7}{8}, \frac{1}{2}-x\right]$	[52,93,112,138,244,285,304,330]
21	$\left[\frac{7}{8}, \frac{7}{8}, x+\frac{3}{4}\right]$	[53,95,111,139,245,287,303,331]
22	$\left[\frac{7}{8}, x+\frac{3}{4}, \frac{7}{8}\right]$	[54,94,155,188,246,286,347,380]
23	$\left[\frac{3}{8}, \frac{1}{2}-x, \frac{7}{8}\right]$	[55,96,156,185,247,288,348,377]
24	$\left[\frac{1}{4}-x, \frac{5}{8}, \frac{5}{8}\right]$	[57,58,74,75,249,250,266,267]
25	$\left[\frac{3}{8}, x+\frac{3}{4}, \frac{3}{8}\right]$	[59,92,150,190,251,284,342,382]
26	$\left[\frac{7}{8}, \frac{1}{2}-x, \frac{3}{8}\right]$	[60,89,151,192,252,281,343,384]
27	$\left[\frac{3}{8}, \frac{3}{8}, x+\frac{3}{4}\right]$	[63,91,101,143,255,283,293,335]
28	$\left[\frac{7}{8}, \frac{3}{8}, \frac{1}{2}-x\right]$	[64,90,100,141,256,282,292,333]
29	$\left[\frac{1}{8}, x+\frac{1}{2}, \frac{5}{8}\right]$	[65,72,79,84,257,264,271,276]
30	$\left[\frac{1}{8}, \frac{5}{8}, x+\frac{1}{2}\right]$	[66,69,76,88,258,261,268,280]
31	$\left[\frac{1}{8}, \frac{5}{8}, \frac{3}{4}-x\right]$	[67,71,77,87,259,263,269,279]
32	$\left[\frac{1}{8}, \frac{3}{4}-x, \frac{5}{8}\right]$	[68,70,78,83,260,262,270,275]
33	$\left[x+\frac{1}{2}, \frac{1}{8}, \frac{5}{8}\right]$	[97,104,133,134,289,296,325,326]
34	$\left[x+\frac{3}{4}, \frac{7}{8}, \frac{7}{8}\right]$	[98,129,147,178,290,321,339,370]
35	$\left[x+\frac{3}{4}, \frac{3}{8}, \frac{3}{8}\right]$	[99,130,146,177,291,322,338,369]
36	$\left[\frac{3}{4}-x, \frac{1}{8}, \frac{5}{8}\right]$	[105,106,122,123,297,298,314,315]
37	$\left[\frac{1}{2}-x, \frac{3}{8}, \frac{7}{8}\right]$	[109,128,158,169,301,320,350,361]
38	$\left[\frac{1}{2}-x, \frac{7}{8}, \frac{3}{8}\right]$	[110,121,157,176,302,313,349,368]
39	$\left[\frac{5}{8}, x, \frac{5}{8}\right]$	[113,120,127,132,305,312,319,324]
40	$\left[\frac{5}{8}, \frac{1}{8}, x+\frac{1}{2}\right]$	[114,117,124,136,306,309,316,328]
41	$\left[\frac{5}{8}, \frac{1}{8}, \frac{3}{4}-x\right]$	[115,119,125,135,307,311,317,327]
42	$\left[\frac{5}{8}, \frac{1}{4}-x, \frac{5}{8}\right]$	[116,118,126,131,308,310,318,323]
43	$\left[x+\frac{1}{2}, \frac{5}{8}, \frac{1}{8}\right]$	[145,152,181,182,337,344,373,374]
44	$\left[\frac{3}{4}-x, \frac{5}{8}, \frac{1}{8}\right]$	[153,154,170,171,345,346,362,363]
45	$\left[\frac{5}{8}, x+\frac{1}{2}, \frac{1}{8}\right]$	[161,168,175,180,353,360,367,372]
46	$\left[\frac{5}{8}, \frac{5}{8}, x\right]$	[162,165,172,184,354,357,364,376]
47	$\left[\frac{5}{8}, \frac{5}{8}, \frac{1}{4}-x\right]$	[163,167,173,183,355,359,365,375]
48	$\left[\frac{5}{8}, \frac{3}{4}-x, \frac{1}{8}\right]$	[164,166,174,179,356,358,366,371]

Table 7: Wyckoff site: 96g, site symmetry: ..m1'

No.	position	mapping
1	$[x, x, z]$	[1,36,193,228]
2	$\left[x+\frac{1}{4}, -z, x+\frac{1}{4}\right]$	[2,43,194,235]
3	$\left[x+\frac{1}{4}, z+\frac{1}{4}, -x\right]$	[3,45,195,237]
4	$\left[z+\frac{1}{4}, x+\frac{1}{4}, -x\right]$	[4,44,196,236]
5	$\left[-z, x+\frac{1}{4}, x+\frac{1}{4}\right]$	[5,46,197,238]
6	$\left[-x, x+\frac{1}{4}, z+\frac{1}{4}\right]$	[6,32,198,224]
7	$\left[x+\frac{1}{4}, -x, z+\frac{1}{4}\right]$	[7,33,199,225]
8	$\left[x, \frac{1}{4}-x, \frac{1}{4}-z\right]$	[8,30,200,222]

continued ...

Table 7

No.	position	mapping
9	$\left[\frac{1}{4}-x,\ x,\ \frac{1}{4}-z\right]$	[9,31,201,223]
10	$\left[\frac{1}{4}-x,\ \frac{1}{4}-x,\ z\right]$	[10,35,202,227]
11	$\left[x+\frac{1}{4},\ x+\frac{1}{4},\ -z\right]$	[11,34,203,226]
12	$[-x,\ -x,\ -z]$	[12,25,204,217]
13	$\left[-x,\ z+\frac{1}{4},\ x+\frac{1}{4}\right]$	[13,47,205,239]
14	$[-x,\ -z,\ -x]$	[14,42,206,234]
15	$\left[z+\frac{1}{4},\ -x,\ x+\frac{1}{4}\right]$	[15,48,207,240]
16	$[-z,\ -x,\ -x]$	[16,41,208,233]
17	$[z,\ x,\ x]$	[17,40,209,232]
18	$[x,\ z,\ x]$	[18,38,210,230]
19	$\left[\frac{1}{4}-x,\ z,\ \frac{1}{4}-x\right]$	[19,26,211,218]
20	$\left[\frac{1}{4}-z,\ \frac{1}{4}-x,\ x\right]$	[20,28,212,220]
21	$\left[\frac{1}{4}-x,\ \frac{1}{4}-z,\ x\right]$	[21,27,213,219]
22	$\left[z,\ \frac{1}{4}-x,\ \frac{1}{4}-x\right]$	[22,29,214,221]
23	$\left[x,\ \frac{1}{4}-z,\ \frac{1}{4}-x\right]$	[23,37,215,229]
24	$\left[\frac{1}{4}-z,\ x,\ \frac{1}{4}-x\right]$	[24,39,216,231]
25	$\left[x,\ x+\frac{1}{2},\ z+\frac{1}{2}\right]$	[49,84,241,276]
26	$\left[x+\frac{1}{4},\ \frac{1}{2}-z,\ x+\frac{3}{4}\right]$	[50,91,242,283]
27	$\left[x+\frac{1}{4},\ z+\frac{3}{4},\ \frac{1}{2}-x\right]$	[51,93,243,285]
28	$\left[z+\frac{1}{4},\ x+\frac{3}{4},\ \frac{1}{2}-x\right]$	[52,92,244,284]
29	$\left[-z,\ x+\frac{3}{4},\ x+\frac{3}{4}\right]$	[53,94,245,286]
30	$\left[-x,\ x+\frac{3}{4},\ z+\frac{3}{4}\right]$	[54,80,246,272]
31	$\left[x+\frac{1}{4},\ \frac{1}{2}-x,\ z+\frac{3}{4}\right]$	[55,81,247,273]
32	$\left[x,\ \frac{3}{4}-x,\ \frac{3}{4}-z\right]$	[56,78,248,270]
33	$\left[\frac{1}{4}-x,\ x+\frac{1}{2},\ \frac{3}{4}-z\right]$	[57,79,249,271]
34	$\left[\frac{1}{4}-x,\ \frac{3}{4}-x,\ z+\frac{1}{2}\right]$	[58,83,250,275]
35	$\left[x+\frac{1}{4},\ x+\frac{3}{4},\ \frac{1}{2}-z\right]$	[59,82,251,274]
36	$\left[-x,\ \frac{1}{2}-x,\ \frac{1}{2}-z\right]$	[60,73,252,265]
37	$\left[-x,\ z+\frac{3}{4},\ x+\frac{3}{4}\right]$	[61,95,253,287]
38	$\left[-x,\ \frac{1}{2}-z,\ \frac{1}{2}-x\right]$	[62,90,254,282]
39	$\left[z+\frac{1}{4},\ \frac{1}{2}-x,\ x+\frac{3}{4}\right]$	[63,96,255,288]
40	$\left[-z,\ \frac{1}{2}-x,\ \frac{1}{2}-x\right]$	[64,89,256,281]
41	$\left[z,\ x+\frac{1}{2},\ x+\frac{1}{2}\right]$	[65,88,257,280]
42	$\left[x,\ z+\frac{1}{2},\ x+\frac{1}{2}\right]$	[66,86,258,278]
43	$\left[\frac{1}{4}-x,\ z+\frac{1}{2},\ \frac{3}{4}-x\right]$	[67,74,259,266]
44	$\left[\frac{1}{4}-z,\ \frac{3}{4}-x,\ x+\frac{1}{2}\right]$	[68,76,260,268]
45	$\left[\frac{1}{4}-x,\ \frac{3}{4}-z,\ x+\frac{1}{2}\right]$	[69,75,261,267]
46	$\left[z,\ \frac{3}{4}-x,\ \frac{3}{4}-x\right]$	[70,77,262,269]
47	$\left[x,\ \frac{3}{4}-z,\ \frac{3}{4}-x\right]$	[71,85,263,277]
48	$\left[\frac{1}{4}-z,\ x+\frac{1}{2},\ \frac{3}{4}-x\right]$	[72,87,264,279]
49	$\left[x+\frac{1}{2},\ x,\ z+\frac{1}{2}\right]$	[97,132,289,324]
50	$\left[x+\frac{3}{4},\ -z,\ x+\frac{3}{4}\right]$	[98,139,290,331]
51	$\left[x+\frac{3}{4},\ z+\frac{1}{4},\ \frac{1}{2}-x\right]$	[99,141,291,333]
52	$\left[z+\frac{3}{4},\ x+\frac{1}{4},\ \frac{1}{2}-x\right]$	[100,140,292,332]
53	$\left[\frac{1}{2}-z,\ x+\frac{1}{4},\ x+\frac{3}{4}\right]$	[101,142,293,334]
54	$\left[\frac{1}{2}-x,\ x+\frac{1}{4},\ z+\frac{3}{4}\right]$	[102,128,294,320]
55	$\left[x+\frac{3}{4},\ -x,\ z+\frac{3}{4}\right]$	[103,129,295,321]

continued ...

Table 7

No.	position	mapping
56	$\left[x+\frac{1}{2},\ \frac{1}{4}-x,\ \frac{3}{4}-z\right]$	[104,126,296,318]
57	$\left[\frac{3}{4}-x,\ x,\ \frac{3}{4}-z\right]$	[105,127,297,319]
58	$\left[\frac{3}{4}-x,\ \frac{1}{4}-x,\ z+\frac{1}{2}\right]$	[106,131,298,323]
59	$\left[x+\frac{3}{4},\ x+\frac{1}{4},\ \frac{1}{2}-z\right]$	[107,130,299,322]
60	$\left[\frac{1}{2}-x,\ -x,\ \frac{1}{2}-z\right]$	[108,121,300,313]
61	$\left[\frac{1}{2}-x,\ z+\frac{1}{4},\ x+\frac{3}{4}\right]$	[109,143,301,335]
62	$\left[\frac{1}{2}-x,\ -z,\ \frac{1}{2}-x\right]$	[110,138,302,330]
63	$\left[z+\frac{3}{4},\ -x,\ x+\frac{3}{4}\right]$	[111,144,303,336]
64	$\left[\frac{1}{2}-z,\ -x,\ \frac{1}{2}-x\right]$	[112,137,304,329]
65	$\left[z+\frac{1}{2},\ x,\ x+\frac{1}{2}\right]$	[113,136,305,328]
66	$\left[x+\frac{1}{2},\ z,\ x+\frac{1}{2}\right]$	[114,134,306,326]
67	$\left[\frac{3}{4}-x,\ z,\ \frac{3}{4}-x\right]$	[115,122,307,314]
68	$\left[\frac{3}{4}-z,\ \frac{1}{4}-x,\ x+\frac{1}{2}\right]$	[116,124,308,316]
69	$\left[\frac{3}{4}-x,\ \frac{1}{4}-z,\ x+\frac{1}{2}\right]$	[117,123,309,315]
70	$\left[z+\frac{1}{2},\ \frac{1}{4}-x,\ \frac{3}{4}-x\right]$	[118,125,310,317]
71	$\left[x+\frac{1}{2},\ \frac{1}{4}-z,\ \frac{3}{4}-x\right]$	[119,133,311,325]
72	$\left[\frac{3}{4}-z,\ x,\ \frac{3}{4}-x\right]$	[120,135,312,327]
73	$\left[x+\frac{1}{2},\ x+\frac{1}{2},\ z\right]$	[145,180,337,372]
74	$\left[x+\frac{3}{4},\ \frac{1}{2}-z,\ x+\frac{1}{4}\right]$	[146,187,338,379]
75	$\left[x+\frac{3}{4},\ z+\frac{3}{4},\ -x\right]$	[147,189,339,381]
76	$\left[z+\frac{3}{4},\ x+\frac{3}{4},\ -x\right]$	[148,188,340,380]
77	$\left[\frac{1}{2}-z,\ x+\frac{3}{4},\ x+\frac{1}{4}\right]$	[149,190,341,382]
78	$\left[\frac{1}{2}-x,\ x+\frac{3}{4},\ z+\frac{1}{4}\right]$	[150,176,342,368]
79	$\left[x+\frac{3}{4},\ \frac{1}{2}-x,\ z+\frac{1}{4}\right]$	[151,177,343,369]
80	$\left[x+\frac{1}{2},\ \frac{3}{4}-x,\ \frac{1}{4}-z\right]$	[152,174,344,366]
81	$\left[\frac{3}{4}-x,\ x+\frac{1}{2},\ \frac{1}{4}-z\right]$	[153,175,345,367]
82	$\left[\frac{3}{4}-x,\ \frac{3}{4}-x,\ z\right]$	[154,179,346,371]
83	$\left[x+\frac{3}{4},\ x+\frac{3}{4},\ -z\right]$	[155,178,347,370]
84	$\left[\frac{1}{2}-x,\ \frac{1}{2}-x,\ -z\right]$	[156,169,348,361]
85	$\left[\frac{1}{2}-x,\ z+\frac{3}{4},\ x+\frac{1}{4}\right]$	[157,191,349,383]
86	$\left[\frac{1}{2}-x,\ \frac{1}{2}-z,\ -x\right]$	[158,186,350,378]
87	$\left[z+\frac{3}{4},\ \frac{1}{2}-x,\ x+\frac{1}{4}\right]$	[159,192,351,384]
88	$\left[\frac{1}{2}-z,\ \frac{1}{2}-x,\ -x\right]$	[160,185,352,377]
89	$\left[z+\frac{1}{2},\ x+\frac{1}{2},\ x\right]$	[161,184,353,376]
90	$\left[x+\frac{1}{2},\ z+\frac{1}{2},\ x\right]$	[162,182,354,374]
91	$\left[\frac{3}{4}-x,\ z+\frac{1}{2},\ \frac{1}{4}-x\right]$	[163,170,355,362]
92	$\left[\frac{3}{4}-z,\ \frac{3}{4}-x,\ x\right]$	[164,172,356,364]
93	$\left[\frac{3}{4}-x,\ \frac{3}{4}-z,\ x\right]$	[165,171,357,363]
94	$\left[z+\frac{1}{2},\ \frac{3}{4}-x,\ \frac{1}{4}-x\right]$	[166,173,358,365]
95	$\left[x+\frac{1}{2},\ \frac{3}{4}-z,\ \frac{1}{4}-x\right]$	[167,181,359,373]
96	$\left[\frac{3}{4}-z,\ x+\frac{1}{2},\ \frac{1}{4}-x\right]$	[168,183,360,375]

Table 8: Wyckoff site: `96h`, site symmetry: `..21'`

No.	position	mapping
1	$[0,\ y,\ -y]$	`[1,14,193,206]`
2	$\left[\frac{1}{4},\ y,\ y+\frac{1}{4}\right]$	`[2,9,194,201]`
3	$\left[\frac{1}{4},\ \frac{1}{4}-y,\ -y\right]$	`[3,10,195,202]`
4	$\left[\frac{1}{4}-y,\ y+\frac{1}{4},\ 0\right]$	`[4,21,196,213]`
5	$\left[y,\ y+\frac{1}{4},\ \frac{1}{4}\right]$	`[5,23,197,215]`
6	$\left[-y,\ \frac{1}{4},\ \frac{1}{4}-y\right]$	`[6,22,198,214]`
7	$\left[y+\frac{1}{4},\ 0,\ \frac{1}{4}-y\right]$	`[7,24,199,216]`
8	$\left[0,\ \frac{1}{4}-y,\ y+\frac{1}{4}\right]$	`[8,13,200,205]`
9	$\left[y+\frac{1}{4},\ \frac{1}{4},\ y\right]$	`[11,20,203,212]`
10	$[-y,\ 0,\ y]$	`[12,17,204,209]`
11	$\left[\frac{1}{4}-y,\ -y,\ \frac{1}{4}\right]$	`[15,19,207,211]`
12	$[y,\ -y,\ 0]$	`[16,18,208,210]`
13	$[0,\ -y,\ y]$	`[25,38,217,230]`
14	$\left[\frac{1}{4},\ -y,\ \frac{1}{4}-y\right]$	`[26,33,218,225]`
15	$\left[\frac{1}{4},\ y+\frac{1}{4},\ y\right]$	`[27,34,219,226]`
16	$\left[y+\frac{1}{4},\ \frac{1}{4}-y,\ 0\right]$	`[28,45,220,237]`
17	$\left[-y,\ \frac{1}{4}-y,\ \frac{1}{4}\right]$	`[29,47,221,239]`
18	$\left[y,\ \frac{1}{4},\ y+\frac{1}{4}\right]$	`[30,46,222,238]`
19	$\left[\frac{1}{4}-y,\ 0,\ y+\frac{1}{4}\right]$	`[31,48,223,240]`
20	$\left[0,\ y+\frac{1}{4},\ \frac{1}{4}-y\right]$	`[32,37,224,229]`
21	$\left[\frac{1}{4}-y,\ \frac{1}{4},\ -y\right]$	`[35,44,227,236]`
22	$[y,\ 0,\ -y]$	`[36,41,228,233]`
23	$\left[y+\frac{1}{4},\ y,\ \frac{1}{4}\right]$	`[39,43,231,235]`
24	$[-y,\ y,\ 0]$	`[40,42,232,234]`
25	$\left[0,\ y+\frac{1}{2},\ \frac{1}{2}-y\right]$	`[49,62,241,254]`
26	$\left[\frac{1}{4},\ y+\frac{1}{2},\ y+\frac{3}{4}\right]$	`[50,57,242,249]`
27	$\left[\frac{1}{4},\ \frac{3}{4}-y,\ \frac{1}{2}-y\right]$	`[51,58,243,250]`
28	$\left[\frac{1}{4}-y,\ y+\frac{3}{4},\ \frac{1}{2}\right]$	`[52,69,244,261]`
29	$\left[y,\ y+\frac{3}{4},\ \frac{3}{4}\right]$	`[53,71,245,263]`
30	$\left[-y,\ \frac{3}{4},\ \frac{3}{4}-y\right]$	`[54,70,246,262]`
31	$\left[y+\frac{1}{4},\ \frac{1}{2},\ \frac{3}{4}-y\right]$	`[55,72,247,264]`
32	$\left[0,\ \frac{3}{4}-y,\ y+\frac{3}{4}\right]$	`[56,61,248,253]`
33	$\left[y+\frac{1}{4},\ \frac{3}{4},\ y+\frac{1}{2}\right]$	`[59,68,251,260]`
34	$\left[-y,\ \frac{1}{2},\ y+\frac{1}{2}\right]$	`[60,65,252,257]`
35	$\left[\frac{1}{4}-y,\ \frac{1}{2}-y,\ \frac{3}{4}\right]$	`[63,67,255,259]`
36	$\left[y,\ \frac{1}{2}-y,\ \frac{1}{2}\right]$	`[64,66,256,258]`
37	$\left[0,\ \frac{1}{2}-y,\ y+\frac{1}{2}\right]$	`[73,86,265,278]`
38	$\left[\frac{1}{4},\ \frac{1}{2}-y,\ \frac{3}{4}-y\right]$	`[74,81,266,273]`
39	$\left[\frac{1}{4},\ y+\frac{3}{4},\ y+\frac{1}{2}\right]$	`[75,82,267,274]`
40	$\left[y+\frac{1}{4},\ \frac{3}{4}-y,\ \frac{1}{2}\right]$	`[76,93,268,285]`
41	$\left[-y,\ \frac{3}{4}-y,\ \frac{3}{4}\right]$	`[77,95,269,287]`
42	$\left[y,\ \frac{3}{4},\ y+\frac{3}{4}\right]$	`[78,94,270,286]`
43	$\left[\frac{1}{4}-y,\ \frac{1}{2},\ y+\frac{3}{4}\right]$	`[79,96,271,288]`
44	$\left[0,\ y+\frac{3}{4},\ \frac{3}{4}-y\right]$	`[80,85,272,277]`
45	$\left[\frac{1}{4}-y,\ \frac{3}{4},\ \frac{1}{2}-y\right]$	`[83,92,275,284]`
46	$\left[y,\ \frac{1}{2},\ \frac{1}{2}-y\right]$	`[84,89,276,281]`

continued ...

Table 8

No.	position	mapping
47	$\left[y+\frac{1}{4},\, y+\frac{1}{2},\, \frac{3}{4}\right]$	[87,91,279,283]
48	$\left[-y,\, y+\frac{1}{2},\, \frac{1}{2}\right]$	[88,90,280,282]
49	$\left[\frac{1}{2},\, y,\, \frac{1}{2}-y\right]$	[97,110,289,302]
50	$\left[\frac{3}{4},\, y,\, y+\frac{3}{4}\right]$	[98,105,290,297]
51	$\left[\frac{3}{4},\, \frac{1}{4}-y,\, \frac{1}{2}-y\right]$	[99,106,291,298]
52	$\left[\frac{3}{4}-y,\, y+\frac{1}{4},\, \frac{1}{2}\right]$	[100,117,292,309]
53	$\left[y+\frac{1}{2},\, y+\frac{1}{4},\, \frac{3}{4}\right]$	[101,119,293,311]
54	$\left[\frac{1}{2}-y,\, \frac{1}{4},\, \frac{3}{4}-y\right]$	[102,118,294,310]
55	$\left[y+\frac{3}{4},\, 0,\, \frac{3}{4}-y\right]$	[103,120,295,312]
56	$\left[\frac{1}{2},\, \frac{1}{4}-y,\, y+\frac{3}{4}\right]$	[104,109,296,301]
57	$\left[y+\frac{3}{4},\, \frac{1}{4},\, y+\frac{1}{2}\right]$	[107,116,299,308]
58	$\left[\frac{1}{2}-y,\, 0,\, y+\frac{1}{2}\right]$	[108,113,300,305]
59	$\left[\frac{3}{4}-y,\, -y,\, \frac{3}{4}\right]$	[111,115,303,307]
60	$\left[y+\frac{1}{2},\, -y,\, \frac{1}{2}\right]$	[112,114,304,306]
61	$\left[\frac{1}{2},\, -y,\, y+\frac{1}{2}\right]$	[121,134,313,326]
62	$\left[\frac{3}{4},\, -y,\, \frac{3}{4}-y\right]$	[122,129,314,321]
63	$\left[\frac{3}{4},\, y+\frac{1}{4},\, y+\frac{1}{2}\right]$	[123,130,315,322]
64	$\left[y+\frac{3}{4},\, \frac{1}{4}-y,\, \frac{1}{2}\right]$	[124,141,316,333]
65	$\left[\frac{1}{2}-y,\, \frac{1}{4}-y,\, \frac{3}{4}\right]$	[125,143,317,335]
66	$\left[y+\frac{1}{2},\, \frac{1}{4},\, y+\frac{3}{4}\right]$	[126,142,318,334]
67	$\left[\frac{3}{4}-y,\, 0,\, y+\frac{3}{4}\right]$	[127,144,319,336]
68	$\left[\frac{1}{2},\, y+\frac{1}{4},\, \frac{3}{4}-y\right]$	[128,133,320,325]
69	$\left[\frac{3}{4}-y,\, \frac{1}{4},\, \frac{1}{2}-y\right]$	[131,140,323,332]
70	$\left[y+\frac{1}{2},\, 0,\, \frac{1}{2}-y\right]$	[132,137,324,329]
71	$\left[y+\frac{3}{4},\, y,\, \frac{3}{4}\right]$	[135,139,327,331]
72	$\left[\frac{1}{2}-y,\, y,\, \frac{1}{2}\right]$	[136,138,328,330]
73	$\left[\frac{1}{2},\, y+\frac{1}{2},\, -y\right]$	[145,158,337,350]
74	$\left[\frac{3}{4},\, y+\frac{1}{2},\, y+\frac{1}{4}\right]$	[146,153,338,345]
75	$\left[\frac{3}{4},\, \frac{3}{4}-y,\, -y\right]$	[147,154,339,346]
76	$\left[\frac{3}{4}-y,\, y+\frac{3}{4},\, 0\right]$	[148,165,340,357]
77	$\left[y+\frac{1}{2},\, y+\frac{3}{4},\, \frac{1}{4}\right]$	[149,167,341,359]
78	$\left[\frac{1}{2}-y,\, \frac{3}{4},\, \frac{1}{4}-y\right]$	[150,166,342,358]
79	$\left[y+\frac{3}{4},\, \frac{1}{2},\, \frac{1}{4}-y\right]$	[151,168,343,360]
80	$\left[\frac{1}{2},\, \frac{3}{4}-y,\, y+\frac{1}{4}\right]$	[152,157,344,349]
81	$\left[y+\frac{3}{4},\, \frac{3}{4},\, y\right]$	[155,164,347,356]
82	$\left[\frac{1}{2}-y,\, \frac{1}{2},\, y\right]$	[156,161,348,353]
83	$\left[\frac{3}{4}-y,\, \frac{1}{2}-y,\, \frac{1}{4}\right]$	[159,163,351,355]
84	$\left[y+\frac{1}{2},\, \frac{1}{2}-y,\, 0\right]$	[160,162,352,354]
85	$\left[\frac{1}{2},\, \frac{1}{2}-y,\, y\right]$	[169,182,361,374]
86	$\left[\frac{3}{4},\, \frac{1}{2}-y,\, \frac{1}{4}-y\right]$	[170,177,362,369]
87	$\left[\frac{3}{4},\, y+\frac{3}{4},\, y\right]$	[171,178,363,370]
88	$\left[y+\frac{3}{4},\, \frac{3}{4}-y,\, 0\right]$	[172,189,364,381]
89	$\left[\frac{1}{2}-y,\, \frac{3}{4}-y,\, \frac{1}{4}\right]$	[173,191,365,383]
90	$\left[y+\frac{1}{2},\, \frac{3}{4},\, y+\frac{1}{4}\right]$	[174,190,366,382]
91	$\left[\frac{3}{4}-y,\, \frac{1}{2},\, y+\frac{1}{4}\right]$	[175,192,367,384]
92	$\left[\frac{1}{2},\, y+\frac{3}{4},\, \frac{1}{4}-y\right]$	[176,181,368,373]
93	$\left[\frac{3}{4}-y,\, \frac{3}{4},\, -y\right]$	[179,188,371,380]

continued ...

Table 8

No.	position	mapping
94	$\left[y+\frac{1}{2},\ \frac{1}{2},\ -y\right]$	`[180,185,372,377]`
95	$\left[y+\frac{3}{4},\ y+\frac{1}{2},\ \frac{1}{4}\right]$	`[183,187,375,379]`
96	$\left[\frac{1}{2}-y,\ y+\frac{1}{2},\ 0\right]$	`[184,186,376,378]`

Table 9: Wyckoff site: `192i`, site symmetry: `11'`

No.	position	mapping
1	$[x,\ y,\ z]$	`[1,193]`
2	$\left[x+\frac{1}{4},\ -z,\ y+\frac{1}{4}\right]$	`[2,194]`
3	$\left[x+\frac{1}{4},\ z+\frac{1}{4},\ -y\right]$	`[3,195]`
4	$\left[z+\frac{1}{4},\ y+\frac{1}{4},\ -x\right]$	`[4,196]`
5	$\left[-z,\ y+\frac{1}{4},\ x+\frac{1}{4}\right]$	`[5,197]`
6	$\left[-y,\ x+\frac{1}{4},\ z+\frac{1}{4}\right]$	`[6,198]`
7	$\left[y+\frac{1}{4},\ -x,\ z+\frac{1}{4}\right]$	`[7,199]`
8	$\left[x,\ \frac{1}{4}-y,\ \frac{1}{4}-z\right]$	`[8,200]`
9	$\left[\frac{1}{4}-x,\ y,\ \frac{1}{4}-z\right]$	`[9,201]`
10	$\left[\frac{1}{4}-x,\ \frac{1}{4}-y,\ z\right]$	`[10,202]`
11	$\left[y+\frac{1}{4},\ x+\frac{1}{4},\ -z\right]$	`[11,203]`
12	$[-y,\ -x,\ -z]$	`[12,204]`
13	$\left[-x,\ z+\frac{1}{4},\ y+\frac{1}{4}\right]$	`[13,205]`
14	$[-x,\ -z,\ -y]$	`[14,206]`
15	$\left[z+\frac{1}{4},\ -y,\ x+\frac{1}{4}\right]$	`[15,207]`
16	$[-z,\ -y,\ -x]$	`[16,208]`
17	$[z,\ x,\ y]$	`[17,209]`
18	$[y,\ z,\ x]$	`[18,210]`
19	$\left[\frac{1}{4}-y,\ z,\ \frac{1}{4}-x\right]$	`[19,211]`
20	$\left[\frac{1}{4}-z,\ \frac{1}{4}-x,\ y\right]$	`[20,212]`
21	$\left[\frac{1}{4}-y,\ \frac{1}{4}-z,\ x\right]$	`[21,213]`
22	$\left[z,\ \frac{1}{4}-x,\ \frac{1}{4}-y\right]$	`[22,214]`
23	$\left[y,\ \frac{1}{4}-z,\ \frac{1}{4}-x\right]$	`[23,215]`
24	$\left[\frac{1}{4}-z,\ x,\ \frac{1}{4}-y\right]$	`[24,216]`
25	$[-x,\ -y,\ -z]$	`[25,217]`
26	$\left[\frac{1}{4}-x,\ z,\ \frac{1}{4}-y\right]$	`[26,218]`
27	$\left[\frac{1}{4}-x,\ \frac{1}{4}-z,\ y\right]$	`[27,219]`
28	$\left[\frac{1}{4}-z,\ \frac{1}{4}-y,\ x\right]$	`[28,220]`
29	$\left[z,\ \frac{1}{4}-y,\ \frac{1}{4}-x\right]$	`[29,221]`
30	$\left[y,\ \frac{1}{4}-x,\ \frac{1}{4}-z\right]$	`[30,222]`
31	$\left[\frac{1}{4}-y,\ x,\ \frac{1}{4}-z\right]$	`[31,223]`
32	$\left[-x,\ y+\frac{1}{4},\ z+\frac{1}{4}\right]$	`[32,224]`
33	$\left[x+\frac{1}{4},\ -y,\ z+\frac{1}{4}\right]$	`[33,225]`
34	$\left[x+\frac{1}{4},\ y+\frac{1}{4},\ -z\right]$	`[34,226]`
35	$\left[\frac{1}{4}-y,\ \frac{1}{4}-x,\ z\right]$	`[35,227]`
36	$[y,\ x,\ z]$	`[36,228]`
37	$\left[x,\ \frac{1}{4}-z,\ \frac{1}{4}-y\right]$	`[37,229]`
38	$[x,\ z,\ y]$	`[38,230]`

continued ...

Table 9

No.	position	mapping
39	$\left[\frac{1}{4}-z,\ y,\ \frac{1}{4}-x\right]$	[39,231]
40	$[z,\ y,\ x]$	[40,232]
41	$[-z,\ -x,\ -y]$	[41,233]
42	$[-y,\ -z,\ -x]$	[42,234]
43	$\left[y+\frac{1}{4},\ -z,\ x+\frac{1}{4}\right]$	[43,235]
44	$\left[z+\frac{1}{4},\ x+\frac{1}{4},\ -y\right]$	[44,236]
45	$\left[y+\frac{1}{4},\ z+\frac{1}{4},\ -x\right]$	[45,237]
46	$\left[-z,\ x+\frac{1}{4},\ y+\frac{1}{4}\right]$	[46,238]
47	$\left[-y,\ z+\frac{1}{4},\ x+\frac{1}{4}\right]$	[47,239]
48	$\left[z+\frac{1}{4},\ -x,\ y+\frac{1}{4}\right]$	[48,240]
49	$\left[x,\ y+\frac{1}{2},\ z+\frac{1}{2}\right]$	[49,241]
50	$\left[x+\frac{1}{4},\ \frac{1}{2}-z,\ y+\frac{3}{4}\right]$	[50,242]
51	$\left[x+\frac{1}{4},\ z+\frac{3}{4},\ \frac{1}{2}-y\right]$	[51,243]
52	$\left[z+\frac{1}{4},\ y+\frac{3}{4},\ \frac{1}{2}-x\right]$	[52,244]
53	$\left[-z,\ y+\frac{3}{4},\ x+\frac{3}{4}\right]$	[53,245]
54	$\left[-y,\ x+\frac{3}{4},\ z+\frac{3}{4}\right]$	[54,246]
55	$\left[y+\frac{1}{4},\ \frac{1}{2}-x,\ z+\frac{3}{4}\right]$	[55,247]
56	$\left[x,\ \frac{3}{4}-y,\ \frac{3}{4}-z\right]$	[56,248]
57	$\left[\frac{1}{4}-x,\ y+\frac{1}{2},\ \frac{3}{4}-z\right]$	[57,249]
58	$\left[\frac{1}{4}-x,\ \frac{3}{4}-y,\ z+\frac{1}{2}\right]$	[58,250]
59	$\left[y+\frac{1}{4},\ x+\frac{3}{4},\ \frac{1}{2}-z\right]$	[59,251]
60	$\left[-y,\ \frac{1}{2}-x,\ \frac{1}{2}-z\right]$	[60,252]
61	$\left[-x,\ z+\frac{3}{4},\ y+\frac{3}{4}\right]$	[61,253]
62	$\left[-x,\ \frac{1}{2}-z,\ \frac{1}{2}-y\right]$	[62,254]
63	$\left[z+\frac{1}{4},\ \frac{1}{2}-y,\ x+\frac{3}{4}\right]$	[63,255]
64	$\left[-z,\ \frac{1}{2}-y,\ \frac{1}{2}-x\right]$	[64,256]
65	$\left[z,\ x+\frac{1}{2},\ y+\frac{1}{2}\right]$	[65,257]
66	$\left[y,\ z+\frac{1}{2},\ x+\frac{1}{2}\right]$	[66,258]
67	$\left[\frac{1}{4}-y,\ z+\frac{1}{2},\ \frac{3}{4}-x\right]$	[67,259]
68	$\left[\frac{1}{4}-z,\ \frac{3}{4}-x,\ y+\frac{1}{2}\right]$	[68,260]
69	$\left[\frac{1}{4}-y,\ \frac{3}{4}-z,\ x+\frac{1}{2}\right]$	[69,261]
70	$\left[z,\ \frac{3}{4}-x,\ \frac{3}{4}-y\right]$	[70,262]
71	$\left[y,\ \frac{3}{4}-z,\ \frac{3}{4}-x\right]$	[71,263]
72	$\left[\frac{1}{4}-z,\ x+\frac{1}{2},\ \frac{3}{4}-y\right]$	[72,264]
73	$\left[-x,\ \frac{1}{2}-y,\ \frac{1}{2}-z\right]$	[73,265]
74	$\left[\frac{1}{4}-x,\ z+\frac{1}{2},\ \frac{3}{4}-y\right]$	[74,266]
75	$\left[\frac{1}{4}-x,\ \frac{3}{4}-z,\ y+\frac{1}{2}\right]$	[75,267]
76	$\left[\frac{1}{4}-z,\ \frac{3}{4}-y,\ x+\frac{1}{2}\right]$	[76,268]
77	$\left[z,\ \frac{3}{4}-y,\ \frac{3}{4}-x\right]$	[77,269]
78	$\left[y,\ \frac{3}{4}-x,\ \frac{3}{4}-z\right]$	[78,270]
79	$\left[\frac{1}{4}-y,\ x+\frac{1}{2},\ \frac{3}{4}-z\right]$	[79,271]
80	$\left[-x,\ y+\frac{3}{4},\ z+\frac{3}{4}\right]$	[80,272]
81	$\left[x+\frac{1}{4},\ \frac{1}{2}-y,\ z+\frac{3}{4}\right]$	[81,273]
82	$\left[x+\frac{1}{4},\ y+\frac{3}{4},\ \frac{1}{2}-z\right]$	[82,274]
83	$\left[\frac{1}{4}-y,\ \frac{3}{4}-x,\ z+\frac{1}{2}\right]$	[83,275]
84	$\left[y,\ x+\frac{1}{2},\ z+\frac{1}{2}\right]$	[84,276]
85	$\left[x,\ \frac{3}{4}-z,\ \frac{3}{4}-y\right]$	[85,277]

continued ...

Table 9

No.	position	mapping
86	$\left[x,\, z+\frac{1}{2},\, y+\frac{1}{2}\right]$	[86,278]
87	$\left[\frac{1}{4}-z,\, y+\frac{1}{2},\, \frac{3}{4}-x\right]$	[87,279]
88	$\left[z,\, y+\frac{1}{2},\, x+\frac{1}{2}\right]$	[88,280]
89	$\left[-z,\, \frac{1}{2}-x,\, \frac{1}{2}-y\right]$	[89,281]
90	$\left[-y,\, \frac{1}{2}-z,\, \frac{1}{2}-x\right]$	[90,282]
91	$\left[y+\frac{1}{4},\, \frac{1}{2}-z,\, x+\frac{3}{4}\right]$	[91,283]
92	$\left[z+\frac{1}{4},\, x+\frac{3}{4},\, \frac{1}{2}-y\right]$	[92,284]
93	$\left[y+\frac{1}{4},\, z+\frac{3}{4},\, \frac{1}{2}-x\right]$	[93,285]
94	$\left[-z,\, x+\frac{3}{4},\, y+\frac{3}{4}\right]$	[94,286]
95	$\left[-y,\, z+\frac{3}{4},\, x+\frac{3}{4}\right]$	[95,287]
96	$\left[z+\frac{1}{4},\, \frac{1}{2}-x,\, y+\frac{3}{4}\right]$	[96,288]
97	$\left[x+\frac{1}{2},\, y,\, z+\frac{1}{2}\right]$	[97,289]
98	$\left[x+\frac{3}{4},\, -z,\, y+\frac{3}{4}\right]$	[98,290]
99	$\left[x+\frac{3}{4},\, z+\frac{1}{4},\, \frac{1}{2}-y\right]$	[99,291]
100	$\left[z+\frac{3}{4},\, y+\frac{1}{4},\, \frac{1}{2}-x\right]$	[100,292]
101	$\left[\frac{1}{2}-z,\, y+\frac{1}{4},\, x+\frac{3}{4}\right]$	[101,293]
102	$\left[\frac{1}{2}-y,\, x+\frac{1}{4},\, z+\frac{3}{4}\right]$	[102,294]
103	$\left[y+\frac{3}{4},\, -x,\, z+\frac{3}{4}\right]$	[103,295]
104	$\left[x+\frac{1}{2},\, \frac{1}{4}-y,\, \frac{3}{4}-z\right]$	[104,296]
105	$\left[\frac{3}{4}-x,\, y,\, \frac{3}{4}-z\right]$	[105,297]
106	$\left[\frac{3}{4}-x,\, \frac{1}{4}-y,\, z+\frac{1}{2}\right]$	[106,298]
107	$\left[y+\frac{3}{4},\, x+\frac{1}{4},\, \frac{1}{2}-z\right]$	[107,299]
108	$\left[\frac{1}{2}-y,\, -x,\, \frac{1}{2}-z\right]$	[108,300]
109	$\left[\frac{1}{2}-x,\, z+\frac{1}{4},\, y+\frac{3}{4}\right]$	[109,301]
110	$\left[\frac{1}{2}-x,\, -z,\, \frac{1}{2}-y\right]$	[110,302]
111	$\left[z+\frac{3}{4},\, -y,\, x+\frac{3}{4}\right]$	[111,303]
112	$\left[\frac{1}{2}-z,\, -y,\, \frac{1}{2}-x\right]$	[112,304]
113	$\left[z+\frac{1}{2},\, x,\, y+\frac{1}{2}\right]$	[113,305]
114	$\left[y+\frac{1}{2},\, z,\, x+\frac{1}{2}\right]$	[114,306]
115	$\left[\frac{3}{4}-y,\, z,\, \frac{3}{4}-x\right]$	[115,307]
116	$\left[\frac{3}{4}-z,\, \frac{1}{4}-x,\, y+\frac{1}{2}\right]$	[116,308]
117	$\left[\frac{3}{4}-y,\, \frac{1}{4}-z,\, x+\frac{1}{2}\right]$	[117,309]
118	$\left[z+\frac{1}{2},\, \frac{1}{4}-x,\, \frac{3}{4}-y\right]$	[118,310]
119	$\left[y+\frac{1}{2},\, \frac{1}{4}-z,\, \frac{3}{4}-x\right]$	[119,311]
120	$\left[\frac{3}{4}-z,\, x,\, \frac{3}{4}-y\right]$	[120,312]
121	$\left[\frac{1}{2}-x,\, -y,\, \frac{1}{2}-z\right]$	[121,313]
122	$\left[\frac{3}{4}-x,\, z,\, \frac{3}{4}-y\right]$	[122,314]
123	$\left[\frac{3}{4}-x,\, \frac{1}{4}-z,\, y+\frac{1}{2}\right]$	[123,315]
124	$\left[\frac{3}{4}-z,\, \frac{1}{4}-y,\, x+\frac{1}{2}\right]$	[124,316]
125	$\left[z+\frac{1}{2},\, \frac{1}{4}-y,\, \frac{3}{4}-x\right]$	[125,317]
126	$\left[y+\frac{1}{2},\, \frac{1}{4}-x,\, \frac{3}{4}-z\right]$	[126,318]
127	$\left[\frac{3}{4}-y,\, x,\, \frac{3}{4}-z\right]$	[127,319]
128	$\left[\frac{1}{2}-x,\, y+\frac{1}{4},\, z+\frac{3}{4}\right]$	[128,320]
129	$\left[x+\frac{3}{4},\, -y,\, z+\frac{3}{4}\right]$	[129,321]
130	$\left[x+\frac{3}{4},\, y+\frac{1}{4},\, \frac{1}{2}-z\right]$	[130,322]
131	$\left[\frac{3}{4}-y,\, \frac{1}{4}-x,\, z+\frac{1}{2}\right]$	[131,323]
132	$\left[y+\frac{1}{2},\, x,\, z+\frac{1}{2}\right]$	[132,324]

continued ...

Table 9

No.	position	mapping
133	$\left[x+\frac{1}{2},\ \frac{1}{4}-z,\ \frac{3}{4}-y\right]$	[133,325]
134	$\left[x+\frac{1}{2},\ z,\ y+\frac{1}{2}\right]$	[134,326]
135	$\left[\frac{3}{4}-z,\ y,\ \frac{3}{4}-x\right]$	[135,327]
136	$\left[z+\frac{1}{2},\ y,\ x+\frac{1}{2}\right]$	[136,328]
137	$\left[\frac{1}{2}-z,\ -x,\ \frac{1}{2}-y\right]$	[137,329]
138	$\left[\frac{1}{2}-y,\ -z,\ \frac{1}{2}-x\right]$	[138,330]
139	$\left[y+\frac{3}{4},\ -z,\ x+\frac{3}{4}\right]$	[139,331]
140	$\left[z+\frac{3}{4},\ x+\frac{1}{4},\ \frac{1}{2}-y\right]$	[140,332]
141	$\left[y+\frac{3}{4},\ z+\frac{1}{4},\ \frac{1}{2}-x\right]$	[141,333]
142	$\left[\frac{1}{2}-z,\ x+\frac{1}{4},\ y+\frac{3}{4}\right]$	[142,334]
143	$\left[\frac{1}{2}-y,\ z+\frac{1}{4},\ x+\frac{3}{4}\right]$	[143,335]
144	$\left[z+\frac{3}{4},\ -x,\ y+\frac{3}{4}\right]$	[144,336]
145	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ z\right]$	[145,337]
146	$\left[x+\frac{3}{4},\ \frac{1}{2}-z,\ y+\frac{1}{4}\right]$	[146,338]
147	$\left[x+\frac{3}{4},\ z+\frac{3}{4},\ -y\right]$	[147,339]
148	$\left[z+\frac{3}{4},\ y+\frac{3}{4},\ -x\right]$	[148,340]
149	$\left[\frac{1}{2}-z,\ y+\frac{3}{4},\ x+\frac{1}{4}\right]$	[149,341]
150	$\left[\frac{1}{2}-y,\ x+\frac{3}{4},\ z+\frac{1}{4}\right]$	[150,342]
151	$\left[y+\frac{3}{4},\ \frac{1}{2}-x,\ z+\frac{1}{4}\right]$	[151,343]
152	$\left[x+\frac{1}{2},\ \frac{3}{4}-y,\ \frac{1}{4}-z\right]$	[152,344]
153	$\left[\frac{3}{4}-x,\ y+\frac{1}{2},\ \frac{1}{4}-z\right]$	[153,345]
154	$\left[\frac{3}{4}-x,\ \frac{3}{4}-y,\ z\right]$	[154,346]
155	$\left[y+\frac{3}{4},\ x+\frac{3}{4},\ -z\right]$	[155,347]
156	$\left[\frac{1}{2}-y,\ \frac{1}{2}-x,\ -z\right]$	[156,348]
157	$\left[\frac{1}{2}-x,\ z+\frac{3}{4},\ y+\frac{1}{4}\right]$	[157,349]
158	$\left[\frac{1}{2}-x,\ \frac{1}{2}-z,\ -y\right]$	[158,350]
159	$\left[z+\frac{3}{4},\ \frac{1}{2}-y,\ x+\frac{1}{4}\right]$	[159,351]
160	$\left[\frac{1}{2}-z,\ \frac{1}{2}-y,\ -x\right]$	[160,352]
161	$\left[z+\frac{1}{2},\ x+\frac{1}{2},\ y\right]$	[161,353]
162	$\left[y+\frac{1}{2},\ z+\frac{1}{2},\ x\right]$	[162,354]
163	$\left[\frac{3}{4}-y,\ z+\frac{1}{2},\ \frac{1}{4}-x\right]$	[163,355]
164	$\left[\frac{3}{4}-z,\ \frac{3}{4}-x,\ y\right]$	[164,356]
165	$\left[\frac{3}{4}-y,\ \frac{3}{4}-z,\ x\right]$	[165,357]
166	$\left[z+\frac{1}{2},\ \frac{3}{4}-x,\ \frac{1}{4}-y\right]$	[166,358]
167	$\left[y+\frac{1}{2},\ \frac{3}{4}-z,\ \frac{1}{4}-x\right]$	[167,359]
168	$\left[\frac{3}{4}-z,\ x+\frac{1}{2},\ \frac{1}{4}-y\right]$	[168,360]
169	$\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ -z\right]$	[169,361]
170	$\left[\frac{3}{4}-x,\ z+\frac{1}{2},\ \frac{1}{4}-y\right]$	[170,362]
171	$\left[\frac{3}{4}-x,\ \frac{3}{4}-z,\ y\right]$	[171,363]
172	$\left[\frac{3}{4}-z,\ \frac{3}{4}-y,\ x\right]$	[172,364]
173	$\left[z+\frac{1}{2},\ \frac{3}{4}-y,\ \frac{1}{4}-x\right]$	[173,365]
174	$\left[y+\frac{1}{2},\ \frac{3}{4}-x,\ \frac{1}{4}-z\right]$	[174,366]
175	$\left[\frac{3}{4}-y,\ x+\frac{1}{2},\ \frac{1}{4}-z\right]$	[175,367]
176	$\left[\frac{1}{2}-x,\ y+\frac{3}{4},\ z+\frac{1}{4}\right]$	[176,368]
177	$\left[x+\frac{3}{4},\ \frac{1}{2}-y,\ z+\frac{1}{4}\right]$	[177,369]
178	$\left[x+\frac{3}{4},\ y+\frac{3}{4},\ -z\right]$	[178,370]
179	$\left[\frac{3}{4}-y,\ \frac{3}{4}-x,\ z\right]$	[179,371]

continued ...

Table 9

No.	position	mapping
180	$\left[y+\frac{1}{2},\, x+\frac{1}{2},\, z\right]$	[180,372]
181	$\left[x+\frac{1}{2},\, \frac{3}{4}-z,\, \frac{1}{4}-y\right]$	[181,373]
182	$\left[x+\frac{1}{2},\, z+\frac{1}{2},\, y\right]$	[182,374]
183	$\left[\frac{3}{4}-z,\, y+\frac{1}{2},\, \frac{1}{4}-x\right]$	[183,375]
184	$\left[z+\frac{1}{2},\, y+\frac{1}{2},\, x\right]$	[184,376]
185	$\left[\frac{1}{2}-z,\, \frac{1}{2}-x,\, -y\right]$	[185,377]
186	$\left[\frac{1}{2}-y,\, \frac{1}{2}-z,\, -x\right]$	[186,378]
187	$\left[y+\frac{3}{4},\, \frac{1}{2}-z,\, x+\frac{1}{4}\right]$	[187,379]
188	$\left[z+\frac{3}{4},\, x+\frac{3}{4},\, -y\right]$	[188,380]
189	$\left[y+\frac{3}{4},\, z+\frac{3}{4},\, -x\right]$	[189,381]
190	$\left[\frac{1}{2}-z,\, x+\frac{3}{4},\, y+\frac{1}{4}\right]$	[190,382]
191	$\left[\frac{1}{2}-y,\, z+\frac{3}{4},\, x+\frac{1}{4}\right]$	[191,383]
192	$\left[z+\frac{3}{4},\, \frac{1}{2}-x,\, y+\frac{1}{4}\right]$	[192,384]